Better Money Labs

Local Credit, Global Settlement: Enhancing Trust-Based Credit Creation with On-Chain Reserve Contracts

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Abstract

In this work, we propose Basis, a monetary framework where both local trust-based (and IOU-based) currencies as well as blockchain-based trust-minimized global ones can coexist via smart contract powered reserves needed when there is no trust (possibly, connecting local trading circles). Coordination with blockchain is needed only during debt note redemption against a reserve contract. Payments (debt note transfers) are done offchain, so with low fees and no need to use low-security centralized blockchain-like systems with high throughput. For coordination and transparency of debt in the system, a tracker service is used, which can not steal money from reserves. We provide analysis of trust-minimization in regards with tracker activities.

Current offchain payment systems backed by on-chain reserves (Lightning Network, Cashu, Fedimint) require full backing and so do not allow for credit creation. In contrast, Basis allows for credit creation with no requirements set for reserves at the protocol level. Then it can be used for community trading without using blockchain (and possibly, without Internet even, just by using a local mesh network), but when trust is insufficient to expand credit, on-chain reserves can be established and used. We expect that on a local scale trust-based economic relationships would dominate, and on a global scale (e.g. when one community is doing trades with another) full coverage with reserves would be needed in most cases. The whole system will use limited in disconnected from real-world needs supply blockchain assets only where they are needed (there is no trust), while enabling monetary expansion whenever possible (so where peer-to-peer trust can be established), thus allowing to create a viable alternative to political money.

The main payment unit is IOU note, which is signed by the issuer and the tracker. Our design allows for debt transferability, if issuer agrees on that, so if peer A issued debt to peer B, and peer B pays with it, fully or partially, to peer C, with A also co-signing, and then A owes C. We show that this design allows for minimal trust to tracker service. A tracker service is committing its state on the blockchain periodically. If a tracker service ceases to exist, it is possible to redeem debt notes against on-chain reserves using this committed state. There could be multiple trackers around the world. We consider different designs, from just centralized server, to federated control, to rollups and sidechains.

We provide implementation of reserve contract as well as offchain clients (tracker server and example clients). We show an example of group trading over mesh network with occasional Internet connection. Another example shows AI agent economies where autonomous agents create credit relationships for services.

Keywords: digital currencies, cryptocurrencies, blockchain, peer-to-peer money, trust-based credit, IOU notes

1 Introduction

Peer-to-peer monetary systems, such as ChainCash [1], enable decentralized (peer-to-peer) money creation through trust and collateral. While on-chain protocols provide transparency and censorship resistance, they often struggle with scalability and cost when every transaction must be recorded on a public blockchain. Off-chain alternatives, such as payment channels and Layer 2 solutions, improve scalability but introduce new trust assumptions, and also do not allow for expansion on top of blockchain assets.

By expansion, we mean creating new monetary instruments (such as credit, IOUs, or representative tokens) that exceed the value of the underlying locked blockchain collateral. This is needed because the emission schedules of decentralized blockchain assets like Bitcoin or Ergo are algorithmically predetermined and entirely disconnected from real-world economic demand. Their supply cannot expand or contract in response to productive activity, trade volumes, or credit needs, making them inherently inelastic. Centralized alternatives like USDT or USDC reintroduce the very trust assumptions that blockchain technology sought to eliminate, requiring users to rely on opaque reserves, auditing processes, and issuer solvency. There is need to bridge this gap by enabling elastic credit creation backed by trust, while still anchoring to on-chain reserves when trust is insufficient.

In this work, we propose Basis, a monetary framework designed for peer-to-peer credit creation and settlement. Basis enables local trust-based currencies to coexist with blockchain-backed global reserves, maintaining an off-chain AVL tree of debt relationships between economic agents, witnessing new notes by signing them, and periodically committing state digests to a public blockchain. This design allows for efficient note circulation off-chain while preserving the security properties of on-chain redemption.

The paper is organized as follows. In Section 2 we describe the design of Basis. Section 3 analyzes security properties. Section 4 provides implementation details. In Section 5 we provide analysis of advantages and drawbacks.

2 Design

We consider a set of economic agents $a_1, ..., a_n$ participating in a peer-to-peer monetary system. Each agent may issue IOU notes, creating debt relationships with other agents. Note acceptance is always optional and at the discretion of the counterparty.

2.1 Note Creation

When an agent (the payer) wishes to create a debt obligation to another agent (the payee), it creates an IOU note containing the payer's signature, the payee's public key, the debt amount, and a timestamp. The note may optionally reference a specific on-chain reserve contract that backs the debt. Before the payee accepts the note, it may request witnessing from a Basis tracker. The tracker verifies the note, updates its local state to record the new debt relationship, and returns a tracker signature certifying that the debt is now tracked. The payee accepts the witnessed note as a valid claim against the payer.

2.2 Note Redemption

A payee holding a witnessed IOU note may redeem it against the payer's reserve at any time. To redeem, the payee presents the note along with both the payer's and tracker's signatures to the reserve contract. The contract verifies the signatures, checks the debt amount against the tracker's committed state, and releases the corresponding collateral to the payee. If the reserve has insufficient collateral, the payee may still hold the note as a claim and attempt redemption later, or the payer may be required to top up the reserve.

In the case where the tracker becomes unavailable, an emergency redemption path exists: after a timeout period (e.g., 3 days or 2160 blocks), the tracker signature becomes optional. The payee can redeem using only the payer's signature and the last committed tracker state from the blockchain, preventing funds from being locked indefinitely.

2.3 Debt Transfer

When an agent wishes to transfer a portion of its debt to a new creditor, the process resembles triangular trade. Suppose peer A owes peer B, and peer B wishes to transfer part of this debt to peer C. Peer B requests peer A to co-sign new notes: one reducing the debt A→B and another creating debt A→C. The tracker witnesses both the reduction in the old debt and the creation of the new debt, and the old note is cancelled. After the transfer, peer A owes peer C directly. This enables debt restructuring and circular trade without requiring full redemption and re-issuance.

2.4 Tracker State

Basis tracks all debt relationships in an AVL tree maintained by the tracker. The tree maps a composite key to the current debt state:

$$\text{Blake2b256}(\text{payerKey}||\text{payeeKey}) \to (\text{totalDebt}, \text{timestamp})$$

Periodically, the tracker commits the root digest of this AVL tree to the public blockchain. This commitment provides a timestamped checkpoint of the global debt state and enables emergency redemption when the tracker is unavailable.

3 Security Analysis

3.1 Tracker Cannot Steal Funds

The tracker only certifies inclusion of notes in its state. It cannot unilaterally create or redeem notes. The issuer's signature is always required for redemption, ensuring that the tracker cannot steal funds.

3.2 Emergency Exit

If the tracker becomes unavailable, an emergency exit mechanism activates after a timeout period (e.g., 3 days or 2160 blocks). After this period, the tracker signature becomes optional for redemption. The same message format is used: $\text{key}||\text{totalDebt}||\text{timestamp}$. Replay attacks are prevented by timestamp verification, and the reserve owner's signature is still required.

3.3 Anti-Censorship

Previously witnessed notes can still be redeemed even if the tracker refuses to witness new notes. The last committed state on the blockchain provides a floor for redemption rights.

4 Implementation

We provide implementation of the Basis protocol in software, including smart contracts for on-chain reserves, a tracker server for coordinating off-chain state, and client libraries for issuing and managing IOU notes.

4.1 Reserve Contract

The reserve contract is implemented on the Ergo blockchain using the ErgoScript smart contract language. It locks collateral (ERG or custom tokens) and allows redemption when presented with a valid debt note and signatures. The contract enforces the timeout rules for emergency redemption without tracker signatures.

4.2 Tracker Server

The tracker server maintains the AVL tree of debt relationships, witnesses new notes by signing them, and periodically commits state digests to the blockchain. It exposes APIs for note witnessing, state queries, and commitment retrieval. The server publishes alerts via the NOSTR protocol for collateralization monitoring.

4.3 Client Libraries

Client libraries provide functionality for agents to issue IOU notes, request tracker witnessing, transfer debt to new creditors, and redeem notes against reserves. The libraries handle signature generation, note serialization, and communication with tracker and blockchain nodes.

4.4 Mesh Network Trading

We demonstrate community trading over mesh networks using Meshtastic LoRa devices. In this scenario, Alice and Bob trade in a village without Internet connectivity. Alice's device creates an IOU note for goods purchased, which is propagated via mesh radio (Bluetooth/WiFi Direct/LoRa) to Bob's device. A local tracker running on a community server signs the IOU, updating the debt ledger. Multiple transactions accumulate off-chain, and when Internet connectivity becomes available through a gateway node, Bob aggregates the IOUs and submits a batch redemption transaction to the Ergo blockchain. This demonstrates offline credit-based trading without forced collateralization, with blockchain used only for final settlement.

Key features demonstrated include: offline transaction creation via mesh propagation, local tracker signature and debt ledger maintenance, batch settlement to reduce blockchain fees, and credit-based trading without collateral requirements.

4.5 AI Agent Credit Creation

We demonstrate autonomous credit creation among AI agents in a software development workflow. A Repo Agent posts a development task with a budget, a Dev Agent completes the work and submits a pull request, and a Test Agent reviews and tests the code. The Repo

Agent creates an IOU note (e.g., 10 ERG) backed by its reserve, which the Dev Agent accepts. The Dev Agent then transfers a portion (e.g., 3 ERG) to the Test Agent for testing services. All agents have Basis wallets with credit limits based on their reputation and payment history. Humans can provide reserve backing as economic feedback, but the agents negotiate terms, create IOUs, and settle debts autonomously without human intervention for each transaction.

This demonstrates: autonomous task negotiation and IOU creation, multi-agent payment splitting, credit limits based on agent reputation, and human reserve backing as optional economic feedback rather than required collateral.

5 Advantages and Drawbacks

- Scalable off-chain tracking with on-chain security guarantees
- Emergency exit prevents lock-in to a single tracker
- Anti-censorship protections for previously witnessed notes
- Periodic commitments enable global state verification

References

[1] Alexander Chepurnoy. Chaincash: Money creation with elastic supply via trust and blockchain assets. `https://github.com/ChainCashLabs/chaincash`. Accessed: 2024-01-01.

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